

4. (2.75 points)

An insurer writes high deductible workers compensation policies with a per-occurrence deductible of \$2,000,000.

Given the following information as of December 31, 2018:

Accident Year	Cumulative Reported Losses (\$000) Limited to \$1,000,000 as of (months)			
	12	24	36	48
2015	450	1,250	2,500	3,300
2016	350	1,500	2,900	
2017	375	1,400		
2018	325			

Accident Year	Limited Expected Values (\$000) at \$500,000 Basic Limit as of (months)			
	12	24	36	48
2015	99	266	356	385
2016	114	285	370	397
2017	109	282	367	394
2018	118	294	376	402

	Limited Expected Values (\$000) at \$1,000,000 Limit as of (months)			
	12	24	36	48
	100	330	528	607
	115	367	565	641
	110	360	558	635
	120	385	583	657

Accident Year	Limited Expected Values (\$000) at \$2,000,000 Limit as of (months)			
	12	24	36	48
2015	100	349	652	809
2016	115	397	724	885
2017	110	388	709	871
2018	120	421	759	921

	Unlimited Expected Values (\$000) as of (months)			
	12	24	36	48
	100	350	690	910
	115	400	785	1,035
	110	390	765	1,010
	120	425	835	1,100

- Assume there is no development after 48 months.
- Accident year 2017 reported losses excess of the \$2,000,000 deductible at 24 months are \$50,000.
- Accident year 2017 reported losses limited to the \$2,000,000 deductible at 24 months are \$2,400,000.

<QUESTION 4 CONTINUED ON NEXT PAGE>

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4. (continued)

a. (1.75 points)

Calculate the expected ultimate loss to the insurer for accident year 2017 as of December 31, 2018 using the procedure outlined by Sahasrabuddhe.

b. (1 point)

An exponential claim size model was used to produce the limited expected values in the tables above. Explain how the estimate of the ultimate loss to the insurer would change if a smaller model parameter was used.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 4				
TOTAL POINT VALUE: 2.75			LEARNING OBJECTIVE(S): A4	
SAMPLE ANSWERS				
Part a: 1.75 points				
<u>Sample 1</u>				
Adjusted Reported Loss @ Basic Limit @ 2018 Level				
	@24	@36	@48	
2015:	1114	1780	2186	$1780 = 2500 \times 376 / 528$
2016:	1202	1930		$1930 = 2900 \times 376 / 565$
	24-36	36-48	48-Ult	
LDF	1.602	1.228	1.0	$LDF = (1780 + 1930) / (1114 + 1202) = 1.602$
CDF	1.967	1.228	1.0	$CDF = 1.602 \times 1.228 \times 1.0 = 1.967$
Adjusted CDF				
$1.967 \times [(1010 - 871) / 402] / [(390 - 388) / 294] = 99.98$				
2017 Ultimate XS Loss of \$2M				
$50,000 \times 99.98 = 4,998,969$				
<u>Sample 2</u>				
15: $1250(294/330) = 1114$ $2500(376/528) = 1780$ $3300(402/607) = 2185$				
16: $1500(294/367) = 1202$ $2900(376/565) = 1930$				
	24-36	36-48		
Development Factors:	1.602	1.228		
LDF to Ultimate = $1.602 \times 1.228 = 1.967$				
Ult XS Loss = $50k \times 1.967 \times [(1010/402)/(390/294)] \times [(1 - 871/1010) / (1 - 388/390)] = 4.999M$				
<u>Sample 3</u>				
Basic limit triangle at the latest limit (AY 2018 level)				
	12	24	36	48
2015:	Not used	1114	1780	2186
2016:	Not used	1202	1930	
2015:24 = $1250 \times 294/330 = 1114$ 2015:36 = $2500 \times 376 / 528 = 1780$				
2015:48 = $3300 \times 402/607 = 2186$				
2016:24 = $1500 \times 294/367 = 1202$ 2016:36 = $2900 \times 376 / 565 = 1930$				
CDF = $1.602 \times 1.228 = 1.967$				
$1.602 \times 1.228 \times [(1010 - 871) / 402] / [(390 - 388) / 294] = 1.967 \times 0.346 / 0.007 = 97.226$				
$50,000 \times 97.226 = \$4,861,300 \rightarrow$ Ultimate excess loss for a 2M deductible.				

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Part b: 1 point

Sample Responses

- A smaller model parameter would lead to a smaller difference between the LEVs. If the LEVs are closer, the CDF adjustment will be larger and the ultimate will therefore be larger.
- If a smaller model parameter were used, that would mean the expected unlimited value was smaller, thus less claims would be expected to be capped. The XS LDF would be larger (the denominator would shrink more than the numerator in the adjustment) so the expected ultimate loss would increase.
- If a smaller model parameter was used, then there would be less impact of limits. This would make the limited LEVs closer to the unlimited expected value. The XS losses would be a smaller portion. This would make XS losses more leveraged, increasing the XS CDF, increasing the XS ultimate.

EXAMINER'S REPORT

Candidates were expected to know how to calculate the expected ultimate for a given layer of losses and understand the relationship between model parameters, development patterns and estimated ultimate losses.

Part a

Candidates were expected to calculate the expected ultimate loss using the Sahasrabuddhe approach for an insurer that writes \$2,000,000 high deductible workers compensation policies.

Common mistakes included:

- Not converting the reported losses limited to \$1M to 2018 basic limits when calculating loss development factors.
- Not subtracting the AY 2017 limited expected values at \$2M from the AY 2017 unlimited expected values to calculate the 24-ultimate LDF excess of \$2M.
- Calculating the total ultimate loss and subtracting the limited loss at \$2M to get the insurer's expected ultimate losses.
- Not using the method prescribed by Sahasrabuddhe.

Part b

Candidates were expected to understand how a change to the claim size model parameter would affect the limited expected values and impact the estimated ultimate loss to an insurer writing high deductible workers compensation policies.

Common mistakes included:

- Not knowing that a smaller parameter of an exponential will cause a lower percentage of losses eliminated and the relationship between the limited expected values and the unlimited expected values changes.
- Not considering the impact that the changing limited expected values has to the LDF adjustment factor. Specifically, the change at earlier maturities of the limited expected value at \$2M to the unlimited expected value is smaller than at later maturities.